AgriPulse

Hyper-Local Demand Intelligence + Satellite Crop Monitoring

Project Specification Document

Theme: Food Security

Stack: React + Telegram Bot + Gemini AI + Google Earth Engine + Firebase

Data Policy

NO hardcoded data. NO mock data. ALL data must be real and live.

1. Problem Statement

Malaysian smallholder farmers operate with near-zero visibility into what their local communities actually need. They plant based on habit or hearsay, disconnected from the real demand of restaurants, warungs, and retailers just kilometers away. The result: chronic overproduction of crops nobody ordered and underproduction of what's urgently needed — a structural failure in food security.

At the same time, these farmers lack access to precision agriculture tools that could help them monitor crop health proactively. Satellite-based vegetation monitoring (NDVI), soil moisture analysis, and drought risk detection exist — but they're locked behind expensive drone hardware and IoT sensor systems that smallholders simply cannot afford. A farmer in Serdang has no way to know that the southern section of their field is under water stress until the damage is visible to the naked eye, by which time it's often too late.

Meanwhile, on the buyer side, restaurants and grocery retailers in the Klang Valley waste hours sourcing inconsistent supply through fragmented informal networks, with no structured channel to signal their recurring needs to nearby farms.

> AgriPulse bridges all three gaps in one platform: a Telegram bot that captures real, structured demand from local buyers; a satellite monitoring system that democratizes precision agriculture using free Sentinel-2 imagery via Google Earth Engine; and a transparent AI engine that synthesizes demand signals, live Malaysian weather data, satellite crop health, and global market conditions into explainable, actionable planting recommendations — so farmers know exactly what to plant, when to plant it, and why.

2. Solution Overview

Side	Interface	Purpose
Consumer / Buyer	Telegram Bot	Declare recurring produce needs (demand signals)
Farmer	React Web App	AI-powered planting recommendations + satellite crop monitoring

The AI engine sits in the middle — ingesting micro-demand (Telegram orders), macro-signals (MET Malaysia weather, news, global markets), and satellite imagery to deliver transparent, explainable recommendations to farmers.

3. Authentication

> Skipped. No login, no sign-up, no Firebase Auth.

Interface	Identity Strategy

Telegram Bot	Telegram userId serves as the unique identifier (auto-provided by Telegram API)
Farmer Web App	Simple onboarding form (name, phone, area) stored in Firestore. No password. Farmer identified by generated farmerId in localStorage

4. Feature 1 — Hyper-Local AI Demand Forecaster

4.1 Consumer Side (Telegram Bot)

Collect structured demand signals from restaurants, warung owners, home cooks, and retailers in a frictionless conversational flow.

Conversational Flow

Stage 1 — Onboarding: User selects their role: restaurant/warung owner, home cook, or grocery retailer.

Stage 2 — Location: User shares GPS location via Telegram or types area manually (e.g., “Serdang, Selangor”).

Stage 3 — Demand Input: User inputs items one by one: “Kangkung 50kg”, “Cili padi 20kg”. AI normalizes crop names. /done to finish.

Stage 4 — Frequency: User selects delivery frequency: weekly, biweekly, or monthly.

Stage 5 — Confirmation: Bot displays full demand profile for review. User can /update anytime.

Bot Commands

Command	Action
/start	Begin onboarding flow
/update	Modify demand profile
/status	View current demand profile
/surplus	Check available surplus from nearby farmers
/lang	Switch language (BM / EN / Chinese)

Notification System

The bot proactively notifies consumers when:

- A nearby farmer lists surplus matching their demand profile
- Price drops occur on items they need
- Seasonal availability changes (e.g., “Durian season starting — 3 farms near you”)

4.2 Farmer Side — AI Demand Dashboard

Demand Heatmap Panel

Aggregated demand by crop within the farmer’s radius (e.g., 30km). Visual breakdown showing buyer count, total kg needed, and demand trend (rising / stable / falling) over the past 4 weeks.

AI Planting Recommendation Card

Top 3 recommended crops to plant now. Each recommendation includes the full Glass-Box AI Breakdown (see Section 6).

Demand-Supply Gap Indicator

Shows undersupplied crops in the area as opportunities. Example: “Cili padi demand is 340kg/week but local supply is only 120kg/week — OPPORTUNITY.”

Data Sources Feeding the AI Engine

Source	Type	What It Provides
Telegram Bot Demand	Micro (local)	Real-time buyer needs, quantities, frequency, location
data.gov.my Weather API	Macro (weather)	7-day MET Malaysia forecast by district, weather warnings
OpenWeatherMap API	Macro (climate)	Long-range climate patterns (El Niño/La Niña) — fallback only
News API	Macro (global)	Import disruptions, crop disease outbreaks, price shocks
Google Earth Engine	Satellite	NDVI, NDWI from Sentinel-2 for the farmer’s own farm

5. Feature 2 — Remote Satellite Crop Monitoring

5.1 Overview

Democratize precision agriculture using free satellite imagery instead of expensive drones/IoT sensors. Farmers draw their farm boundary on a map and receive automated health reports.

5.2 Farm Registration Flow

Step 1 — Draw Boundary: Google Maps embedded with polygon drawing tool. Farmer clicks to draw farm boundary (min 3 points). Auto-calculate farm area in hectares.

Step 2 — Crop Declaration: What are you currently growing? (dropdown + free text). When did you plant? (date picker). Expected harvest date.

Step 3 — Monitoring Activated: “Your farm is now being monitored by satellite!” First report within 24 hours. Updates every 5 days (Sentinel-2 revisit cycle).

5.3 Satellite Data Pipeline

1. Farm polygon (GeoJSON) submitted by farmer
2. Google Earth Engine processes: NDVI (vegetation health), NDWI (soil moisture), change detection vs 2 weeks prior
3. Gemini AI analyzes satellite data: generates plain-language health report, risk alerts, actionable recommendations
4. Results displayed on Farmer Dashboard with color-coded map overlay

5.4 Farm Health Dashboard

1. Farm Map View

Satellite overlay of the farmer’s polygon with color-coded NDVI: green zones (healthy), yellow zones (mild stress — needs attention), red zones (critical — possible disease, drought, or nutrient deficiency).

2. Health Score Card

Single score from 0–100 summarizing overall farm health, with a trend arrow: ↑ improving, → stable, ↓ declining.

3. AI Health Report

Natural language analysis in the farmer’s chosen language. Example:

> Laporan Ladang — 15 April 2026 Skor Kesihatan: 74/100 (Baik) Bahagian utara ladang anda sihat. Bahagian selatan menunjukkan tekanan ringan — kemungkinan kekurangan air. Cadangan AI: Bahagian selatan menunjukkan NDVI rendah (0.35 vs purata 0.55). Berdasarkan data cuaca MET Malaysia, tiada hujan dijangka 7 hari. Kami cadangkan pengairan tambahan di zon ini. Sebab cadangan: 1. Data NDVI: 0.35 (bawah ambang sihat 0.45) 2. Ramalan MET Malaysia: Tiada hujan, 7 hari 3. Indeks kelembapan tanah (NDWI): Rendah

4. Historical Timeline

Swipeable timeline showing farm health evolution over weeks and months.

6. Glass-Box Transparent AI

Every AI-generated recommendation across the entire app follows the same explainability pattern. No black boxes. Farmers see exactly WHY.

6.1 Recommendation Card Structure

Each recommendation card displays these sections:

1. PERMINTAAN TEMPATAN (Local Demand) — exact buyer count, kg demand, current supply, supply gap
2. CUACA & IKLIM (Weather & Climate) — MET Malaysia forecast + long-range climate outlook
3. PASARAN GLOBAL (Global Market) — import/export disruptions, local price trends
4. KESIHATAN LADANG (Farm Health) — NDVI value, NDWI value, recommended planting zone
5. UNJURAN KEUNTUNGAN (Profit Forecast) — estimated cost, revenue, and margin
6. KEYAKINAN AI (AI Confidence) — confidence percentage with explanation of uncertainty

6.2 Transparency Rules

1. Every number is sourced — “47 buyers” links to anonymized Telegram demand data
2. Every prediction shows confidence — percentage + reasoning for uncertainty
3. Contradictory signals are shown — if weather says yes but market says no, BOTH are displayed
4. Farmer can override — “AI says plant chili, but I know my soil. Override → Plant tomato”
5. Feedback loop — After harvest, farmer reports actual yield and AI learns and improves

7. Multilingual Architecture

7.1 Zero-Barrier Localization Strategy

Component	Languages	Implementation
Telegram Bot	BM, EN, Chinese (Cantonese)	Auto-detect from first message via Gemini, /lang to switch
React Web App	BM, EN	i18next library with JSON locale files
AI Responses	Matches user’s language	Language param passed to Gemini system prompt

The Telegram bot auto-detects the user’s language from their first message using Gemini, sets the locale accordingly, and all subsequent responses use that language. Users can override anytime with the /lang command.

8. Tech Stack

Layer	Technology	Purpose
Consumer Interface	Telegram Bot API (Node.js)	Demand collection, notifications
Farmer Interface	React 18 + Vite + TailwindCSS + shadcn/ui	Dashboard, maps, monitoring
AI Engine	Gemini API (gemini-2.0-flash)	Demand analysis, crop recommendations, health reports
Database	Firebase Firestore	Demand profiles, farm data, user records
Maps	Google Maps JavaScript API	Farm boundary drawing, demand heatmaps
Satellite	Google Earth Engine API	NDVI, NDWI, change detection
Weather (Primary)	data.gov.my Weather API	7-day MET Malaysia forecast, weather warnings
Weather (Fallback)	OpenWeatherMap API	Long-range climate data (El Niño/La Niña only)
i18n	i18next	Multilingual support (React)
Hosting	Vercel / Firebase Hosting + Cloud Functions	Web app + bot backend

8.1 data.gov.my Weather API

Primary weather data source. No API key required — free public open data from MET Malaysia.

Endpoint	Description	Update Frequency
GET /forecast	7-day general forecast by district	Daily
GET /warning	Active weather warnings (thunderstorms, strong winds)	As needed
GET /warning/earthquake	Earthquake alerts	As needed

Supports district-level filtering. Forecast values are in Bahasa Melayu — native fit for the BM-first UX. Limitation: 7-day forecast only (no long-range predictions). For El Niño/La Niña seasonal patterns, fall back to OpenWeatherMap.

9. Firestore Data Model

buyers/{telegramUserId}

Stores role, location (lat/lng/area_name), language preference, array of demand items (crop, quantity_kg, frequency), and timestamps.

farmers/{farmerId}

Stores name, phone, area_name, language, and array of farms — each with name, boundary (GeoJSON polygon), area_hectares, and current_crops array.

demand_aggregates/{area_name}/{crop}

Auto-computed aggregation: total_weekly_kg, buyer_count, trend (rising/stable/falling), last_updated. Recomputed via Cloud Function on every buyer write.

satellite_reports/{farmerId}/{farmIndex}/{reportDate}

Stores ndvi_average, ndvi_map_url, health_score (0–100), ai_analysis text, alerts array, and generation timestamp.

ai_recommendations/{farmerId}/{timestamp}

Stores recommended_crops array, reasoning object (demand, weather, market, farm_data), confidence score, farmer_override (if any), and actual_yield (filled post-harvest).

10. Real Data Enforcement Rules

These rules are absolute. No exceptions.

Rule	Detail
No mock weather data	All forecasts from api.data.gov.my/weather or OpenWeatherMap live calls
No fake demand numbers	All aggregates computed from real Telegram bot entries in Firestore
No placeholder NDVI	Satellite data fetched from Google Earth Engine for actual drawn polygon
No hardcoded crop lists	Crop names normalized by Gemini from real buyer input
No dummy buyer profiles	Every buyer record created through actual Telegram bot onboarding
No simulated recommendations	Every AI recommendation generated by Gemini using real data
Loading states required	Show “No data yet — waiting for buyers” instead of fake numbers
Empty states designed	Every panel has meaningful empty state with call to action

Two interfaces, one mission:

Close the gap between what Malaysia needs and what Malaysia grows.